

Discussion 6 Worksheet — *Answer Key*

DS 120 Staff

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Problem 1

Given

$$\mathbf{u} = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 1 \\ 4 \\ -2 \end{bmatrix},$$

the projection of $\mathbf{u}$ onto $\mathbf{v}$ is

$$\text{proj}_{\mathbf{v}}(\mathbf{u}) = \frac{\mathbf{v} \cdot \mathbf{u}}{\mathbf{v} \cdot \mathbf{v}} \mathbf{v}.$$

Compute the dot products:

$$\mathbf{v} \cdot \mathbf{u} = (1)(2) + (4)(-1) + (-2)(3) = 2 - 4 - 6 = -8, \quad \mathbf{v} \cdot \mathbf{v} = 1^2 + 4^2 + (-2)^2 = 1 + 16 + 4 = 21.$$

Hence

$$\text{proj}_{\mathbf{v}}(\mathbf{u}) = \frac{-8}{21} \begin{bmatrix} 1 \\ 4 \\ -2 \end{bmatrix} = \begin{bmatrix} -\frac{8}{21} \\ -\frac{32}{21} \\ \frac{16}{21} \end{bmatrix}.$$

Problem 2

Project $\mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ onto $\mathcal{S} = \text{span}\{\mathbf{a}_1, \mathbf{a}_2\}$ where

$$\mathbf{a}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{a}_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad A = [\mathbf{a}_1 \quad \mathbf{a}_2].$$

Use $P = A(A^\top A)^{-1}A^\top$:

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad A^\top A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \quad (A^\top A)^{-1} = \frac{1}{3} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}.$$

Compute $A^\top \mathbf{b} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$ and

$$\hat{\mathbf{b}} = A(A^\top A)^{-1}A^\top \mathbf{b} = A \left(\frac{1}{3} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \end{bmatrix} \right) = A \left(\frac{1}{3} \begin{bmatrix} 5 \\ 2 \end{bmatrix} \right) = \begin{bmatrix} \frac{7}{3} \\ \frac{2}{3} \\ \frac{5}{3} \end{bmatrix}.$$

So

$$\boxed{\text{proj}_{\mathcal{S}}(\mathbf{b}) = \begin{bmatrix} \frac{7}{3} \\ \frac{2}{3} \\ \frac{5}{3} \end{bmatrix}}.$$

Problem 3

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ with $ad - bc \neq 0$.

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

Then

$$A^{-1}A = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \frac{1}{ad - bc} \begin{bmatrix} da - bc & db - bd \\ -ca + ac & -cb + ad \end{bmatrix} = \frac{1}{ad - bc} \begin{bmatrix} ad - bc & 0 \\ 0 & ad - bc \end{bmatrix} = I.$$

Thus $\boxed{A^{-1}A = I}$.

Problem 4

Model:

$$y = m_1x_1 + m_2x_2 + m_3x_3 + b, \quad \mathbf{m} = \begin{bmatrix} m_1 \\ m_2 \\ m_3 \\ b \end{bmatrix}.$$

Using rows $[x_1 \ x_2 \ x_3 \ 1]$, the system $A\mathbf{m} = \mathbf{y}$ is

$$A = \begin{bmatrix} 1 & 1 & 200 & 1 \\ 0 & 1 & 50 & 1 \\ 3 & 2 & 5 & 1 \\ 2 & 1 & 10 & 1 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} 1500 \\ 600 \\ 7000 \\ 3200 \end{bmatrix}.$$

Solving $(A^T A)^{-1} A^T \mathbf{y}$ gives

$\mathbf{m} = \begin{bmatrix} 1252.94 \\ 2535.29 \\ -2.3529 \\ -1817.65 \end{bmatrix} \Rightarrow m_1 = 1252.94, m_2 = 2535.29, m_3 = -2.3529, b = -1817.65.$

Prediction for $x_1 = 2, x_2 = 1, x_3 = 5$:

$$\hat{y} = 1252.94(2) + 2535.29(1) - 2.3529(5) - 1817.65 = \boxed{3211.76}.$$

Problem 5

We fit $y = ax^2 + bx + c$ through the points $(-3, 4), (0, 0), (4, 4)$. Write the matrix A with columns for $x^2, x, 1$:

$$\begin{bmatrix} 9 & -3 & 1 \\ 0 & 0 & 1 \\ 16 & 4 & 1 \end{bmatrix}.$$

Also the vector y is $\begin{bmatrix} 4 \\ 0 \\ 4 \end{bmatrix}$

Then

$$(A^T A)^{-1} A^T y = \begin{bmatrix} 1/3 \\ -1/3 \\ 0 \end{bmatrix}.$$

So the fitted equation is

$$y = \frac{1}{3}x^2 - \frac{1}{3}x$$